

IBRAHIM ROGERS

MECHANICAL DESIGN | TEST SYSTEMS | HARDWARE DEVELOPMENT

B.S. Mechanical Engineering, Northeastern University · Expected May 2028
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SELECTED WORK

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02	Contamination Root-Cause Investigations	Parker Hannifin · 2026
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Liquid Valve Endurance Tester & Leak-Test Fixture

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Context: Precision fluid-control valves for medical and life-science systems. My work spans test-system design, fixture design, and root-cause investigation on returned product.

My Role: Requirements definition and hydraulic architecture for a closed-loop endurance tester, through to building and commissioning the loop; mechanical design of a supplier-side leak-test fixture.

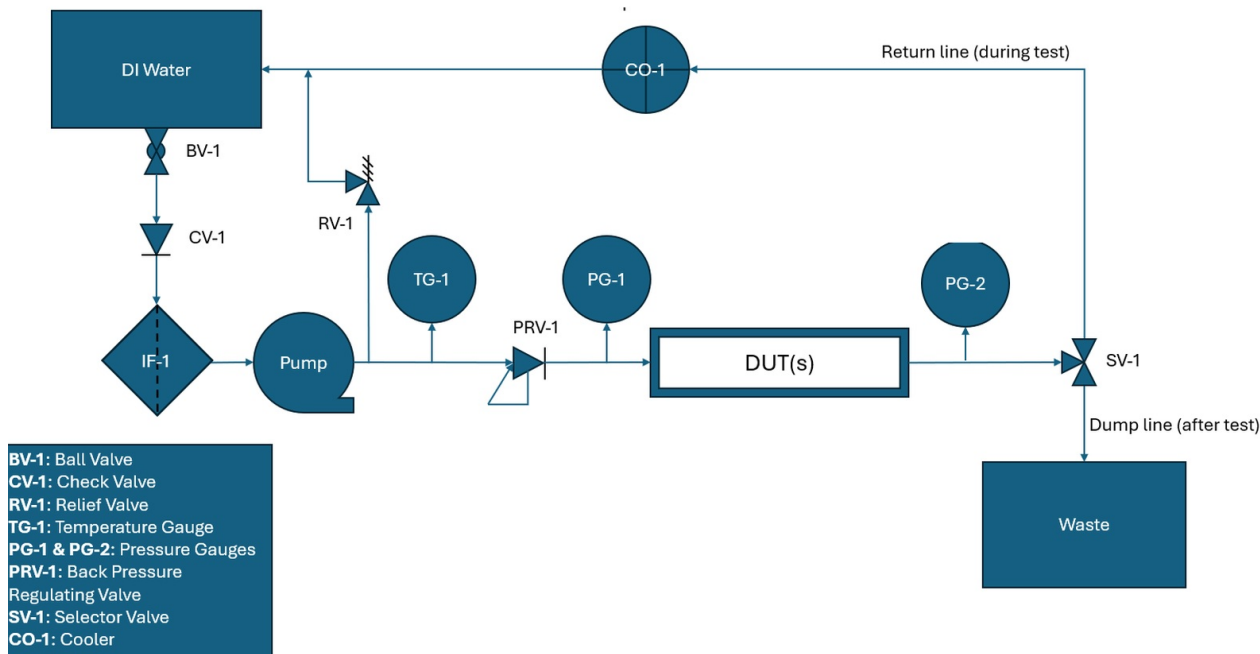
CLOSED-LOOP DI-WATER VALVE ENDURANCE TESTER

Consolidated conflicting inputs from quality, manufacturing, and engineering into a single defined test-system scope, then developed the architecture and produced build-ready drawings, schematics, and P&IDs to match.

Test capacity	Multi-valve bank, cycled simultaneously
Flow per valve / aggregate	Low per-valve flow / bank-level aggregate
Operating pressure	Low hundreds of psig
Fluid temperature control	Tight band around ambient
Working fluid	DI water, continuous recirculation

Sized pump, chiller, reservoir, manifold, and instrumentation, and integrated pressure control, relief protection, and filtration for continuous recirculation. I then built the loop from those drawings: plumbed the suction and return runs, installed filtration and isolation valves, wired and commissioned the pump, and leak-checked the assembled system.

LOOP SCHEMATIC



Test loop P&ID. DI water draws from the reservoir through an isolation valve, check valve, and inline filter to the pump. A relief valve returns to tank; a back-pressure regulating valve sets test pressure upstream of the DUT bank. Gauges sit either side of the DUT so pressure drop can be read across it, and a selector valve routes the outlet either back through the cooler to the reservoir during test or to waste on drain-down.

AS-BUILT HARDWARE



Pump, filter, and valve assembly. Jet pump with accumulator, inline filter housing, and isolation valves plumbed with braided suction and return lines. Built from the schematic above.



Loop routed to reservoir. Return and suction runs from the pump cart to the bulkhead fitting on the reservoir, with the accumulator tank at left. Assembled and leak-checked before instrumentation.

PRESSURE-DECAY LEAK-TEST FIXTURE

Designed a fixture in Autodesk Inventor for supplier-side incoming inspection of a molded fluid-control component.

Leak resolution	Below commodity-instrument resolution
Trapped volume	Small fraction of a cc
Method selected	Pressure decay
Seal architecture	Guarded multi-port
Probe sizing basis	Worst-case tolerance stack + MMC/LMC

DESIGN REASONING

I selected pressure decay over steady-state make-up flow because flow instrumentation fine enough to resolve the required leak rate was not available at acceptable cost. Locating and sealing functions are separated so the probe seals against the orifice without over-constraining a molded part that carries real dimensional variation.

Contamination Root-Cause Investigations

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

OVERVIEW

Led two independent root-cause investigations into recurring return failures on precision fluid-control valves. Each involved a distinct contamination mechanism and a separate population of returned parts.

INVESTIGATION A — ORIFICE CORROSION

Scope	Several-hundred-unit return population
Variables isolated	Supplier, stock, cleaning, floor process
Outcome	Contamination source isolated; mitigation validated

I designed isolation experiments across each candidate variable instead of inspecting finished returns alone. That let the investigation separate a supplier-side cause from a downstream cleaning or floor-process cause.

INVESTIGATION B — FOREIGN-OBJECT CONTAMINATION

Scope	Separate, smaller return population
Method	Teardown and comparative inspection against conforming parts

A separate contamination mechanism from Investigation A. It required its own teardown protocol and comparison baseline; the corrosion investigation's isolation variables did not transfer.

WHAT I TOOK FROM IT

- **Two investigations, not one.** Treating these as a single "contamination" bucket would have averaged two different mechanisms into one wrong answer. Splitting them is what let each get isolated correctly.
- **Comparative inspection needs a real baseline.** Conforming parts had to be inspected alongside the returns to make "different" mean anything.
- **Mitigation validation is a process.** Confirming a fix takes sustained data, not a single passing result.

Packaging Redesign & Machine-Vision Inspection

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

PACKAGING REDESIGN — TRANSIT DAMAGE ON A LOW-VOLUME COMPONENT

Trigger	Field-reported transit damage
Part profile	Low-volume, high-value
Constraint	Fixed cost ceiling set by unit economics

The binding constraint here was economic, not technical. A packaging fix that eliminated damage but cost more than the value it protected would never have shipped, so the design had to converge on something effective enough to stop the transit failure mode while staying inside the cost ceiling.

MACHINE-VISION INSPECTION SYSTEM — IN DEVELOPMENT

Goal	Flag defects on a production cell before shipment
Stack	Python, OpenCV
Status	In development

This extends the computer-vision work from my pan/tilt tracking system (Project 06) into a production application. A production inspection system has to run unattended against real part variation, which is a harder detection problem than tracking a fiducial marker under controlled lighting.

Printed Fixturing — Load-Frame Nest & Compliant Petri Tray

Quality & Sustaining Engineering Co-op — Parker Hannifin, Precision Fluidics · Hollis, NH · May 2026 – Present

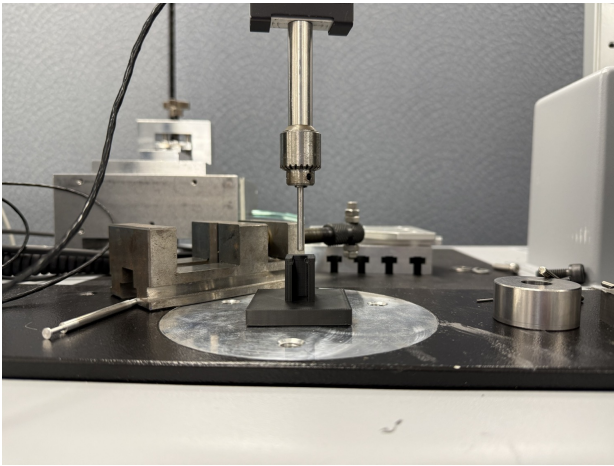
OVERVIEW

Context: Two printed fixtures supporting the same product family — one holds a molded component for mechanical characterization in a load frame, the other holds petri dishes on the line.

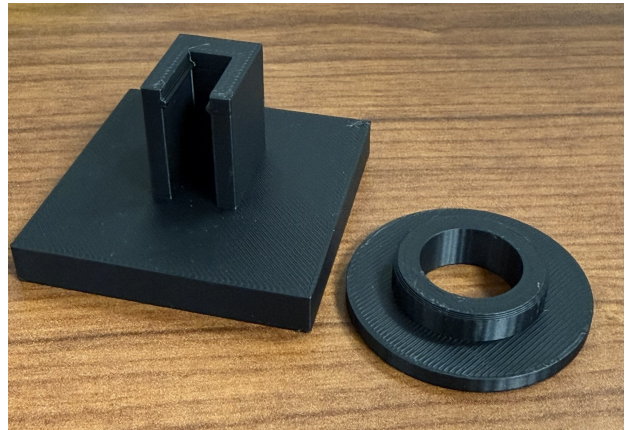
My Role: Feature and mechanism design, CAD, print iteration, and functional checkout on both parts.

LOAD-FRAME NEST

The nest exists so the part loads the same way every time. Hand-positioning a component under a probe introduces run-to-run variation in the same measurement the test is trying to resolve, so the part is constrained laterally by the slot and carried on a base wide enough to sit flat on the platen without rocking. The sealing face stays open so the probe contacts the part and nothing else. The version shown is a printed prototype; the fixture is to be rebuilt in PETG.



Nest in the load frame. Printed nest locating a molded component under the probe during sealing-force characterization. The slot constrains the part laterally while leaving the sealing face open.

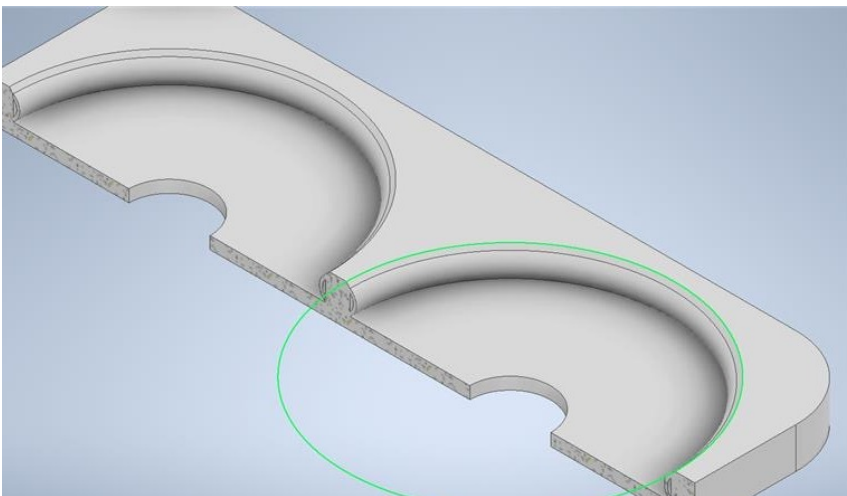


Nest and locating ring, as printed. The slotted nest sits on a wide base that registers flat to the platen. The ring is a separate locating feature printed alongside it.

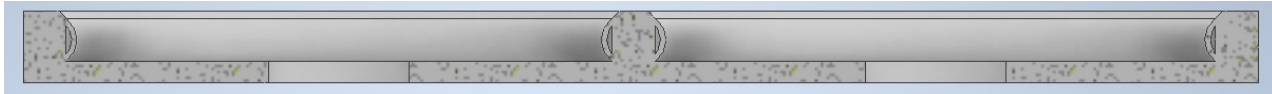
COMPLIANT PETRI TRAY

A dish sitting in a plain recess is held by clearance alone, and that clearance has to be loose enough to accept dish-to-dish variation, which means the dish moves. A clamp or a spring fixes that by adding a part and a loading step. The compliant wall does it with neither: the dish deflects the wall on the way in, and the wall holds it.

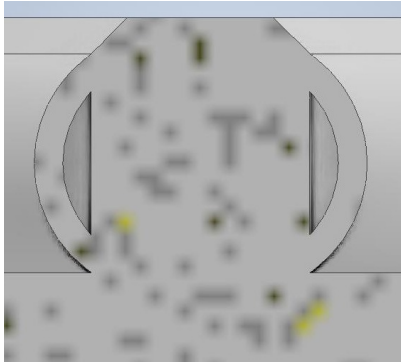
The retention wall arches over a void, with narrow slits either side acting as the hinge, so the material between them rotates rather than the wall bending as a simple cantilever. The void sets how far the wall can travel inward; the hinge sets where it gives.



Two-pocket module. Each pocket carries a compliant retention wall around the recess. Modules butt together so a tray is built from repeated units rather than drawn at a fixed length.



Section through the tray. The retention wall arches over a void running behind it, thinning toward the crown.



Flexure detail. Narrow slits either side of the void act as the hinge, so the material between them rotates rather than the wall bending as a simple cantilever.

FIRST PRINT



First print. The retention wall did not resolve at the thickness it was drawn at, so the pockets came out as plain recesses and dishes sat loose in them.

The first print did not produce the feature. The retention wall was drawn thinner than the printer can resolve, so the pockets came out as plain recesses and dishes sat loose in them. Thickening the neck to a section the process can actually build is what recovers the feature — the compliance then comes from the arch and the slit hinge rather than from a paper-thin wall.

DESIGN REASONING

The thinnest section sits at the crown of the arch, directly over the void and between the two slit tips. That is where stress concentrates and where a failure would start, and on an FDM part it is loaded across layers rather than along them. The only part that exists is the first print, in PLA, in which the wall never resolved; the revised geometry has not been printed yet. Retention across repeated insertions is therefore not established, and PLA is not the material that would establish it — it creeps under sustained deflection. That check waits on a PETG build.

WHAT I TOOK FROM IT

- **A dimension is only real if the process can hold it.** The wall was drawn at a thickness no FDM nozzle can lay down, so the feature existed in CAD and nowhere else.
- **Compliance is geometry, not thinness.** Arch height and the slit hinge set the deflection. Chasing it with a thinner wall is what pushed the feature below the process floor in the first place.
- **Match the claim to the material.** A PLA part can answer whether the geometry grips. It cannot answer whether it still grips after a few hundred insertions.

Refreshable Braille Tactile Display

Hardware Product Lab — Forge, Northeastern University · Jan 2026 – Jun 2026

OVERVIEW

Challenge: Build a refreshable Braille cell that is low-cost, survives thousands of actuation cycles, and is small enough to tile into a line of text. The governing constraint is dimensional: a six-dot cell spans 6.25 mm, and any motor worth using is wider than the cell it drives.

My Role: Mechanism architecture and encoding scheme, CAD, full drawing package, clearance and tolerance-stack iteration, prototype fabrication, and cycle testing.

DESIGN PROGRESSION

The cell went through three drive architectures. Each change was driven by a constraint the previous one could not meet.

1 — Lead screw	A lead screw integrated directly into the key system, chosen to minimize packaging by putting the drive inside the cell envelope. Dropped on time and cost before it was built.
2 — Direct rack & pinion	One motor and pinion driving a single rack beneath the pin block, lifting six dome pins off cam profiles. This is the build that was drawn, fabricated, and cycle-tested.
3 — Bevel geartrain	Two racks per cell driven through a bevel geartrain from outboard motors, with each rack encoding all eight states of its three pins. The architecture the project ended on.

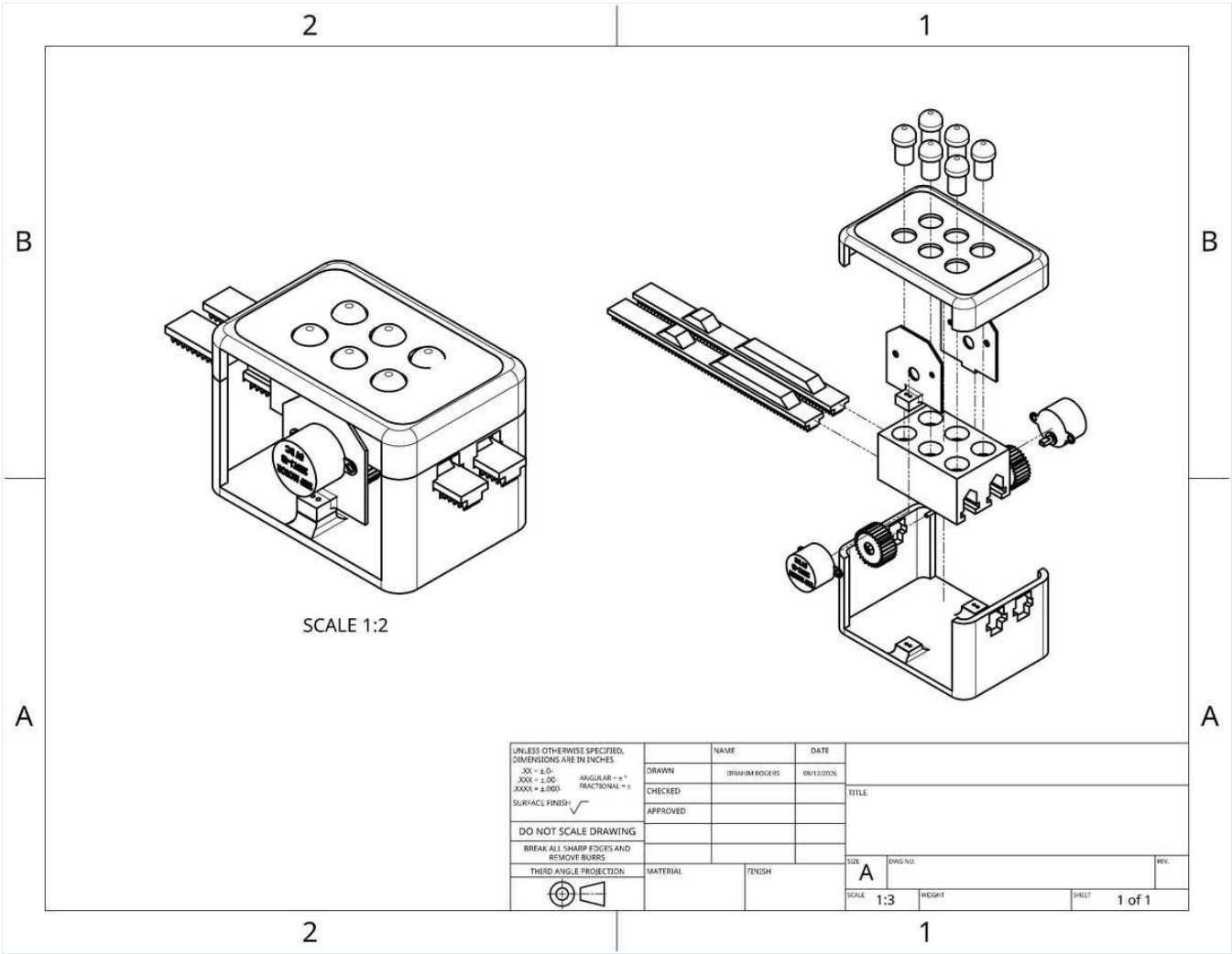
The pivot from the lead screw was a schedule and budget decision rather than a technical one, and it cost packaging density. The move from direct rack and pinion to the bevel geartrain was the opposite: it was made specifically to solve tiling, which the direct-drive layout could not do at reading pitch.

The pages that follow run in that order — the direct-drive build and its drawing package first, then the bevel geartrain architecture and the at-size hardware.

ARCHITECTURE 2 — DIRECT RACK & PINION

After the lead-screw concept was dropped, the cell was built as a direct-drive mechanism at enlarged scale: one motor and pinion advancing a single rack beneath the pin block, with each dot carried on its own dome pin. This is the version that was released as a drawing package, fabricated, and cycled. The drawings, tolerance work, and failure analysis on the next three pages all come from it.

FULL ASSEMBLY — EXPLODED & ISOMETRIC



Top-level assembly drawing. Left: assembled enclosure (Scale 1:2). Right: exploded view (Scale 1:3) showing the six dome pins, letter-housing insert, guide bracket, pin-guide block, drive motor and pinion, rack, and enclosure top/bottom. Self-authored release drawing, dimensioned in inches per standard third-angle projection.

[illegible]

2

1

Ø.354

R.075

R.100

1.100

.200

.160

SECTION A - A

.500

2

1

1 of 1

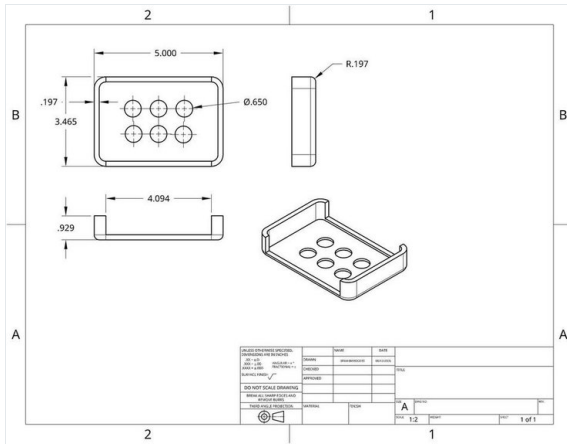
Technical drawing of a mechanical part. The drawing shows a horizontal rectangular plate with a length dimension of 1,400. A circular hole with a diameter of $\varnothing 100$ is located on the right side. A diagonal rod or pin is shown below the plate. The drawing is framed by a coordinate system with labels 1, 2, A, and B.

DO NOT SCALE DRAWING
 DIMENSIONS TO CENTER UNLESS OTHERWISE NOTED
 DIMENSIONS TO SURFACE UNLESS OTHERWISE NOTED

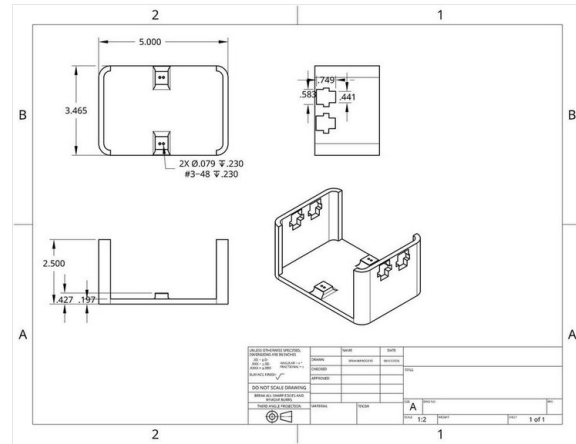
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Guide dowel. Ø.100 in × 1.400 in ground pin. Locates the pin-guide block against the enclosure and keeps rack travel aligned under side load.

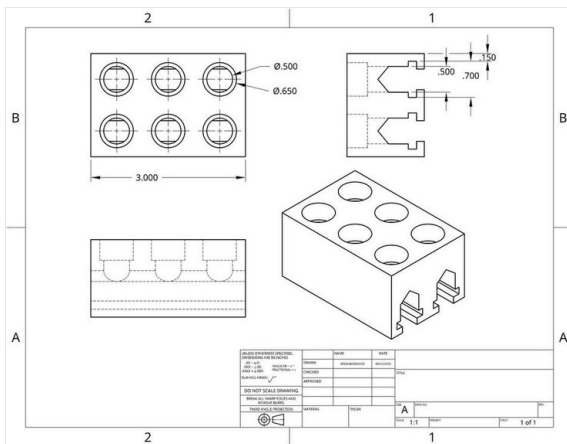
ENCLOSURE & GUIDE STRUCTURE



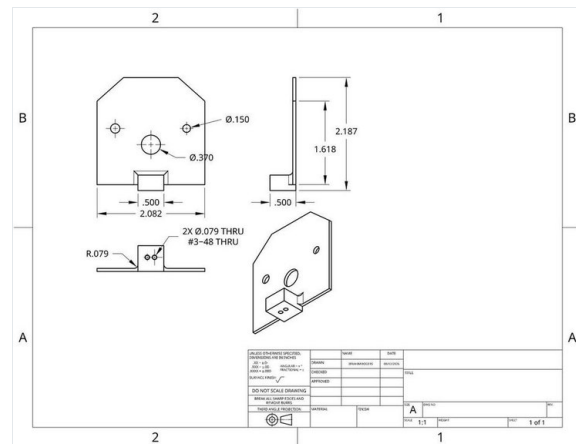
Enclosure top cover. 5.000 × 3.465 in with six Ø.650 in access holes for the dome pins and R.197 in corner fillets. Mounting flange at .929/4.094 in locates the cover to the bottom chassis.



Enclosure bottom chassis. Matching 5.000 × 3.465 × 2.500 in shell with keyed mounting tabs (2x Ø.079 in thru, #3-48 thru) for the pin-guide block and motor mount.



Pin-guide block. 3.000 in wide manifold with six bores (Ø.500 in through, Ø.650 in counterbore) carrying the dome pins, and snap-fit tabs (.150/.500/.700 in) that key it to the enclosure.



Motor-side bracket. L-bracket locating the drive motor relative to the pin-guide block, with Ø.370 in motor-shaft clearance and two Ø.150 in mounting holes on a .500 in tab.

These drawings belong to the enlarged six-pin build. Its clearance strategy does not carry over directly to the at-size cell, which uses printed features far finer than anything dimensioned here.

KEY PROBLEM — TOLERANCE STACK-UP UNDER CYCLIC LOADING

Early prototypes jammed when actuated under side load. Failure tracking isolated two compounding causes:

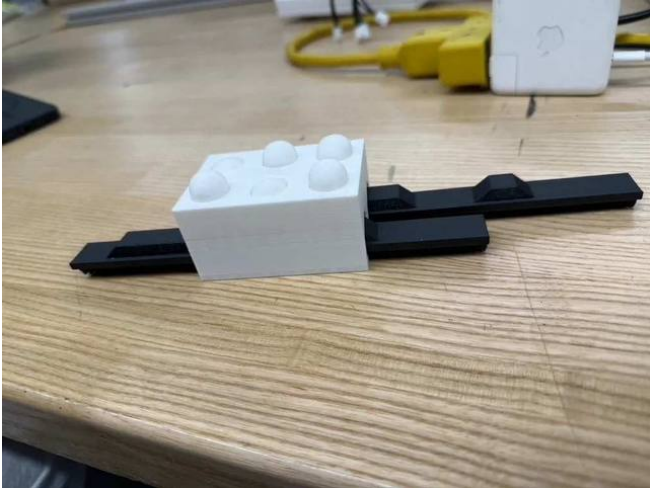
- Guide-channel clearance too tight at 0.1 mm nominal, leaving insufficient margin once print tolerance and side deflection stacked.
- Dimensional creep in PLA after sustained cycling, drifting clearances further out of spec.

Fix: opened guide clearance to 0.15 mm, added lead-in chamfers, and moved to PETG. **Result:** jam rate fell from roughly 15% to under 2% across 2,000+ test cycles. The released drawing package on the preceding pages reflects this fix directly, including the Ø.500 in pin bore sized to hold that clearance under load.

WHAT I TOOK FROM IT

- **Clearance is not just a manufacturing number.** It has to absorb print tolerance and accumulated wear, not only nominal fit.
- **Material choice is a lifecycle decision.** PLA was fine for form checks and wrong for cyclic loading, and that only showed up in testing.
- **Track failures, don't guess.** Logging jam events by condition pointed at side-load binding specifically, which made the fix targeted.
- **Design for assembly.** Consolidating cam and guide features cut part count from 12 to 8 and simplified the assembly sequence.
- **Cross-domain constraint.** Worked with the electrical side to keep actuation force inside servo torque limits while still meeting tactile force requirements.

PHYSICAL PROTOTYPE



Assembled prototype — side view. White housing carries the pin actuation mechanism and rides on the black guide rails, which provide lateral constraint. Shows rail engagement and the travel available to the rack.



Front three-quarter view. Shows the six-dot face and the housing-to-rail interface where side-load binding originated. Lead-in chamfers on the rails were added at this interface to ease pin entry.

This PLA/PETG prototype predates the released drawing package on the preceding pages and is the build that surfaced the side-load binding described there.

ARCHITECTURE 3 — BEVEL GEARTRAIN & POSITIONAL ENCODING

Direct drive put the motor in line with the cell, which set a minimum pitch far wider than a reader expects and made tiling impossible. The final architecture moves the motors outboard and drives each cell through a bevel geartrain, then uses the two resulting racks to encode the whole character instead of lifting dots one at a time.

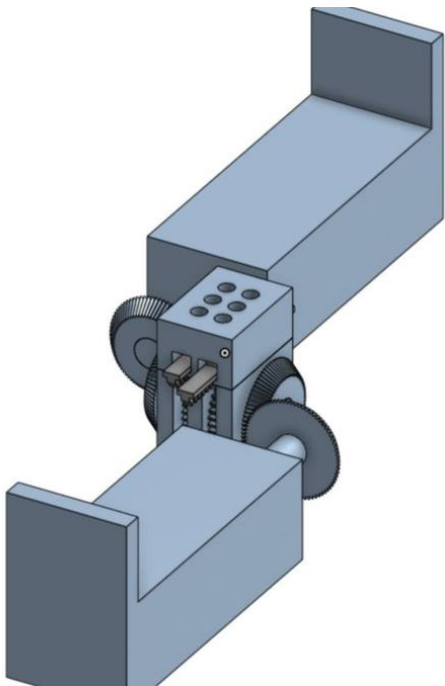
POSITIONAL ENCODING — 64 CHARACTERS, TWO MOTORS

The obvious architecture gives every dot its own actuator, six per cell. Instead, each cell carries two racks. Each rack is a stepped cam that encodes all eight raised/lowered combinations for its own set of three pins, and the motor drives the rack to whichever position the character calls for.

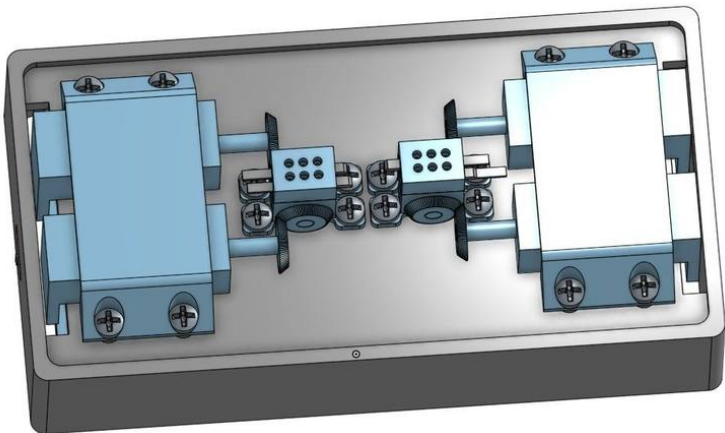
Rack position	1	2	3	4	5	6	7	8
Pins raised	—	a	b	c	ab	ac	bc	abc

Rack A carries dots 1–2–3, rack B carries dots 4–5–6. Eight positions against eight positions is 64 states, which is exactly the six-dot Braille set including the blank cell. Actuator count per cell drops from six to two.

CELL ARCHITECTURE & TILING



Single cell. The two racks enter from opposite faces, each driven by its own bevel gear. Rack channels extend away from the cell in opposite directions, which keeps the gearing and motors clear of the six-dot footprint.



Two-cell assembly. Cells sit adjacent at reading pitch with motors outboard and racks driving inward. Each cell is independently addressed, so a character can be rewritten without disturbing its neighbor.

Motors outboard and racks entering from opposite faces is what makes the cell tileable. Putting the actuator behind the dots would set a minimum cell pitch far wider than the 6.25 mm a reader expects, so the drive train had to leave the footprint entirely.



Production-scale cell, printed parts, and drive motor. The cell body sits on the output shaft of the gearmotor that drives it, with a rack, bevel gear, and dome pins alongside. Everything is shown at true relative scale, which is the point of the photograph: the drive hardware is larger than the cell it actuates.

WHY THIS WAS THE HARD PART

Actuating six dots is not difficult on its own. Actuating them inside 6.25 mm, repeatedly, with hardware that can sit next to copies of itself, is the entire problem. The encoding scheme and the outboard drive layout are both answers to that constraint, not to the actuation.

Vision-Guided Pan/Tilt Tracking System

Independent Project · Dec 2025 – Present

OVERVIEW

Goal: Build a closed-loop target tracking platform — mechanical gimbal, electronics, and control software — that holds a marker centered in frame and recovers on its own when the target is lost.

System: Raspberry Pi 5, camera module, PCA9685 servo driver, dual-servo pan/tilt gimbal, OpenCV ArUco detection, Python control logic, Flask live video stream.



Pan/tilt gimbal. Custom two-axis 3D-printed housing with direct-drive servo coupling. No gear train, which removes backlash from the positioning path.



Electronics integration. Raspberry Pi 5 driving the PCA9685 servo controller over I²C, with servo power on a separate rail from compute power to keep actuation transients off the Pi.

MECHANICAL DESIGN DECISIONS

- **Backlash:** direct-drive servo coupling with a press-fit interface, eliminating gear-train slop from the control loop.
- **Center of mass:** camera placement chosen to reduce servo torque load and improve response during fast target movement.
- **Rigidity:** gimbal frame reinforced to limit flex under rapid acceleration so the camera stays aligned during dynamic tracking.
- **Cable routing:** integrated to avoid wire interference across the full pan/tilt range.

INTEGRATION DEBUG — CAMERA NOT ENUMERATING

PROBLEM

No frame acquisition and no video feed. From the outside the whole system looked dead, with no indication whether the fault was mechanical, electrical, or software.

PROCESS

- Used `raspistill` to separate camera hardware faults from software faults.
- Verified I²C device enumeration to confirm the Pi was seeing devices on the bus.
- Found improper CSI ribbon seating compounded by outdated camera firmware — two independent faults presenting as one symptom.
- Updated firmware and added error handling in Python for graceful frame-drop recovery.

LESSON

Integration bugs concentrate at interface boundaries. Isolating with diagnostic tools beat guessing at the failure location, especially with two faults stacked.

CONTROL IMPLEMENTATION

- PID loop on pan and tilt axes to smooth servo motion and suppress oscillation around the target; gains tuned empirically on hardware.
- Automated re-acquisition via spiral search when the marker leaves frame or is occluded.
- Error handling for dropped frames, I²C communication failures, and servo limit violations.

WHAT I TOOK FROM IT

- **Stage the integration.** Camera alone, then servos alone, then detection, then full loop. Bringing it up in stages saved hours over debugging everything at once.
- **Textbook PID isn't enough.** Real servo dynamics, friction, and loop latency meant the gains had to come from testing.
- **Cross-domain debugging is the actual skill.** Mechanical slop, power distribution, and software timing all produced similar-looking tracking errors.

Gamma Shielding Test System

Co-Founder & COO — Melagen Labs, VC-backed aerospace materials startup · Jan 2024 – Sep 2024

OVERVIEW

Challenge: Validate the shielding effectiveness of a melanin-HDPE composite against an aluminum reference using gamma radiation, with limited lab equipment and a startup budget.

My Role: Co-designed the test fixture, developed the experimental protocol for repeatable measurement, ran the testing, and performed the statistical analysis.

Key Result: Attenuation comparable to aluminum at roughly one-third the density, supported by ANOVA.

TEST SYSTEM & SETUP



Complete test system. Fixed-geometry fixture holds a repeatable Cs-137 source-to-detector distance. A modular sample holder allows quick swaps between composite thicknesses without disturbing the measurement geometry.

DESIGN REQUIREMENTS

- **Repeatability:** fixed source-to-detector positioning to minimize geometric variation between runs, the dominant error source in this measurement.
- **Modularity:** quick-change sample mounting to test multiple thicknesses and compositions efficiently.
- **Safety:** Cs-137 source containment and shielding per lab safety protocol.
- **Low-cost fabrication:** built from lab-available materials within startup budget.

SAMPLE PREPARATION & MEASUREMENT



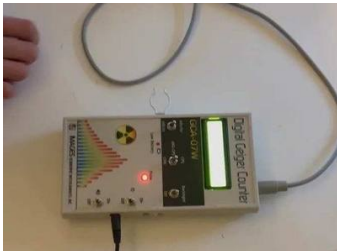
Material preparation. Melanin powder measured to a controlled mass so concentration stayed consistent batch to batch.



Sample fabrication. Composite cast in aluminum pans for consistent thickness, under controlled material-handling protocol.



Test specimens. Composite plates at varying thickness, measured and tested against aluminum references.



Radiation measurement. Calibrated Geiger counter at fixed distance from the shielded source, logging count rate for each shielding condition against an aluminum baseline.

ENGINEERING PROCESS

- **Fixture design for repeatability:** hand-positioning produced unacceptable run-to-run measurement scatter. Adding fixed mechanical alignment features cut that variation substantially and was the single biggest improvement to data quality.
- **Rapid iteration under constraints:** when the first fixture drifted after repeated sample changes, redesigned the mounting within days using available shop tools.
- **Statistical validation:** ANOVA across material types and thicknesses, with regression modeling of shielding effectiveness against melanin concentration.
- **Uncertainty analysis:** documented contributions from source activity calibration, detector efficiency, and geometric positioning so the results held up to outside review.
- **Technical communication:** translated the data into engineering presentations used in fundraising.

WHAT I TOOK FROM IT

- **Fixture design drives data quality.** In experimental work, repeatability is the whole game. Small mechanical improvements to alignment had far more impact on confidence in the result than anything downstream in the analysis.
- **Constraints force decisions.** Limited budget and equipment meant choosing what actually mattered to measure well.

Technical Skills

Tools and methods I can speak to in detail and have applied on the work in this portfolio

CAD & DESIGN

Autodesk Inventor · SolidWorks · Onshape · AutoCAD
GD&T (MMC/LMC) · worst-case tolerance stack-up · interference analysis · fixture design · engineering drawings · P&ID development

PROGRAMMING & ANALYSIS

Python (OpenCV, machine-vision inspection, pandas) · C++ · MATLAB · Excel
ANOVA / regression · I²C and PWM interfaces · Raspberry Pi · Arduino · hardware–software integration

QUALITY & VALIDATION

Failure analysis · root-cause investigation · FMEA · dimensional inspection and metrology · experimental test design · nonconformance investigation · validation and test documentation

FLUID & TEST SYSTEMS

Hydraulic/pneumatic architecture · pressure regulation and relief · closed-loop recirculation · pump and chiller sizing · manifold design · pressure-decay and flow-based leak test methods · test-stand architecture

MANUFACTURING & PROTOTYPING

FDM 3D printing (PLA, PETG) · SLA/resin printing · rapid prototyping · mechanical assembly · fixture fabrication · design for manufacturing · design for assembly

ENGINEERING COMMUNICATION

Failure-analysis reporting · design reviews · cross-functional requirements definition · vendor and supplier coordination · technical presentations to non-technical audiences

ADDITIONAL PROJECT EXPERIENCE

Assistive Sweeper Robot

Cornerstone II, London · Jan–Apr 2025

Co-designed an RC sweeper in SolidWorks, working drivetrain geometry against motor torque and ground clearance constraints. Iterated through a rapid prototyping sprint for operator usability and assembly.

Autonomous Maze Robot

Cornerstone I, Oakland · Sep–Dec 2024

Programmed sensor-based autonomous navigation logic. Diagnosed drivetrain binding and electrical contact failures through systematic root-cause analysis across a two-week test cycle.

Hydrogen Fuel Cell RC Vehicle

TKS Innovation Fellow · 2023–24

Designed a hydrogen fuel cell power system, managing power distribution and integration. Won a Google-sponsored pitch competition for the technical proposal.